

Vendor Repair Agreement

VRA-2025-005

Document ID: VRA-2025-005

Effective Date: April 1, 2021 (original); Addendum A appended March 10, 2026

Term: Initial 5-year warranty + 5-year Extended Service Contract through March 31, 2031

Agreement Type: OEM Warranty and Extended Service Contract

Parties: Cascadia Renewable Operations LLC ("Operator") and Siemens Gamesa Renewable Energy A/S ("Siemens Gamesa" or "OEM"), a Danish corporation with U.S. service operations headquartered in Orlando, Florida.

Governing Documents: Siemens Gamesa Wind Turbine Service Manual, Revision 4.2 (June 2024) — VTM-SIEMENS-GAMESA-4-2; Operator Standard MS-2025-03 (in conflict; see Addendum A).

Distribution: Procurement; CAISO Operations; NW-PowerPool Operations; Engineering; Internal Audit; Siemens Gamesa Service Director.

Preamble

This Agreement establishes Operator's OEM service relationship with Siemens Gamesa for the Siemens Gamesa-manufactured wind turbines in Operator's fleet, comprising the initial five-year OEM warranty (commissioned in 2021) and a continuous five-year Extended Service Contract through March 2031.

Assets Covered. This Agreement applies to the following Siemens Gamesa-manufactured wind turbines:

Asset Tag	Site	Region
WIN-10130	CAISO-WIND-026	CAISO
WIN-10054	CAISO-WIND-026	CAISO
WIN-10131	NW-PowerPool-WIND-024	NW-PowerPool
WIN-10011	NW-PowerPool-WIND-024	NW-PowerPool

These four units account for a disproportionate share of Operator's FY2025 forced-outage MWh and emergency-invoice spend (see OIR-2025-001, OIR-2025-003, OIR-2025-007).

Section 1 — Scope of Services

1.1 OEM Service Scope. OEM shall provide:

- (a) Major-component diagnostic, repair, and replacement services for the covered turbines, performed in accordance with Siemens Gamesa Wind Turbine Service Manual, Revision 4.2 (VTM-SIEMENS-GAMESA-4-2);
- (b) Genuine Siemens Gamesa replacement parts and OEM-certified consumables;
- (c) OEM warranty administration for the initial 5-year warranty period and the Extended Service Contract;
- (d) Coordination with Operator's preventive partner (Cascadia Power Services Inc. — see VRA-2025-001) for routine non-major-component work that does not require OEM presence.

1.2 Subcontract Authority. OEM service technicians may be subcontracted to Apex Emergency Grid Solutions under VRA-2025-002 §1.3 when emergency mobilization is required inside the EPA-2024-11 four-hour window.

1.3 Procedural Authority. OEM service work performed under this Agreement is governed by VTM-SIEMENS-GAMESA-4-2. **Addendum A acknowledges a known conflict between VTM-SIEMENS-GAMESA-4-2 and Operator Standard MS-2025-03 with respect to gearbox vibration replacement thresholds.** See Addendum A and MOC-2025-022.

Section 2 — Rate Schedule

2.1 OEM Labor Rate. \$310.00 per OEM technician-hour for in-warranty work (covered at no cost to Operator for warranty-eligible defects); \$385.00 per OEM technician-hour for out-of-warranty work and Extended Service Contract work.

2.2 Premium Multipliers.

- (a) Weekend / holiday: 1.45× base rate;
- (b) Subcontracted-emergency activation (through VRA-2025-002): 1.8× pass-through to Apex, plus Apex's coordination labor.

2.3 OEM Component Replacement Pricing (selected items, OEM catalog Rev. Jan-2026).

Component	OEM Catalog Price (per unit)
Main gearbox (full replacement assembly)	\$1,200,000
High-speed shaft bearing kit	\$96,500
Yaw drive motor (full)	\$58,000

Component	OEM Catalog Price (per unit)
Generator (full replacement, 3.0 MW class)	\$740,000
Pitch system controller (full module)	\$34,500
Blade root bolt kit (set, per blade)	\$6,200

All replacement components are priced at OEM catalog; Operator is not permitted under this Agreement to substitute non-OEM parts on covered units without forfeiting Extended Service Contract coverage on the affected component.

2.4 Annual Escalation. OEM catalog escalation per Siemens Gamesa's standard catalog revision cycle. Labor escalation at CPI-U (National), capped at 4.5%.

Section 3 — Service Level Agreements

3.1 Major-Component Repair Mobilization. OEM technicians on site within 72 hours of formal Operator request for major-component diagnostic or repair.

3.2 Emergency Mobilization. Where the EPA-2024-11 four-hour window applies, mobilization is via subcontract through VRA-2025-002 (Apex).

3.3 Parts Lead Times. Major-component parts lead times per OEM catalog. Gearbox assemblies: 14–20 weeks. Generator assemblies: 12–18 weeks. Bearing kits: 4–8 weeks.

Section 4 — Approval & Authorization Workflow

4.1 Routine Warranty Claims. OEM warranty claims for covered defects are processed directly between OEM and Operator's Regional Maintenance Planner.

4.2 Replacement-Class Recommendations. Any OEM recommendation for replacement of a major component (gearbox, generator, main bearing, blade) must be reviewed against the procedural-conflict mechanism in Section 4.3 and authorized through a Repair Approval Memo (RAM).

4.3 Procedural Conflict Mechanism (Addendum A). Where the OEM's recommendation is based on a VTM-SIEMENS-GAMESA-4-2 threshold that conflicts with Operator Standard MS-2025-03, the recommendation shall be flagged in writing, routed to Operator's Engineering function under MOC-2025-022, and held pending Engineering disposition before any capital is committed. The most material example as of the date of this revision is the gearbox vibration replacement threshold (see Addendum A §A.1).

4.4 EPA-2024-11 Interaction. Where OEM service is delivered via subcontract under VRA-2025-002, the activation authorization travels through the EPA-2024-11 process, but the procedural-conflict mechanism in §4.3 continues to apply.

Section 5 — Performance Metrics

5.1 Tracked Metrics.

- (a) Warranty claim turnaround (target: 30 days from claim to disposition);
- (b) OEM parts lead-time adherence;
- (c) Repeat-failure rate on OEM-replaced components within 24 months (a metric introduced in Addendum A).

5.2 Note on Repeat Failures. Operator's internal tracking shows multiple repeat forced-outage events on WIN-10131 and WIN-10011 following OEM-recommended interventions. See OIR-2025-007 and RCA-2025-004. The pattern is currently under Engineering review and is not yet the subject of formal warranty dispute.

Section 6 — Limitations, Exclusions, Termination

6.1 Exclusions. Non-Siemens-Gamesa assets; consequential damages; lost generation revenue; environmental remediation.

6.2 Non-OEM Parts Forfeiture. Use of non-OEM parts on a covered unit voids Extended Service Contract coverage on the affected component.

6.3 Termination. OEM may terminate Extended Service Contract on 365 days' notice. Operator may terminate on 180 days' notice. Initial warranty period is non-terminable.

6.4 Document Hierarchy (Within OEM Scope). Where work is performed under this Agreement, VTM-SIEMENS-GAMESA-4-2 is the OEM's procedural authority. However, per Addendum A, **Operator Standard MS-2025-03 governs the decision to authorize replacement-class capital expenditure** even where the OEM's manual would mandate replacement at a lower threshold. The mechanism for routing such conflicts is MOC-2025-022.

Addendum A — Acknowledged Procedural Conflict (Appended March 10, 2026)

A.1 Gearbox Vibration Replacement Threshold. Section 4 of VTM-SIEMENS-GAMESA-4-2 establishes a gearbox-replacement criterion at a measured vibration of

3.5 mm/s RMS (running mean, broadband). Operator Standard MS-2025-03, §5.4 establishes the corresponding internal threshold at **5.0 mm/s RMS**, with refurbishment-first disposition between 5.0 and 7.0 mm/s RMS, and full replacement only at 7.0 mm/s RMS or above (or upon confirmed metallurgical evidence of imminent failure).

The conflict is acknowledged and is the subject of MOC-2025-022 ("Management of Change Request — Override Siemens Gamesa OEM Gearbox Vibration Replacement Threshold for Out-of-Warranty Units"), submitted March 2025 and currently in Engineering review as of the date of this Addendum.

A.2 Practical Effect. Until MOC-2025-022 is closed, OEM recommendations to replace gearboxes at measured vibration levels between 3.5 and 5.0 mm/s RMS will be flagged under §4.3 above and held for Operator Engineering disposition. Two such cases — WIN-10131 (October 2025) and WIN-10011 (December 2025) — are pending under RAM-2025-005 and RAM-2025-007 respectively.

A.3 No Waiver. Nothing in this Addendum waives any OEM warranty obligation or operator capital-replacement obligation. The Addendum addresses only the *procedural routing* of replacement-class recommendations.

Closing

Authorized Signatures

For Cascadia Renewable Operations LLC:

VP, Asset Management — original signed April 1, 2021; Addendum A countersigned March 10, 2026.

For Siemens Gamesa Renewable Energy A/S (U.S. Service):

Vice President, Service Americas — original signed April 1, 2021; Addendum A countersigned March 10, 2026.

Attachments:

- Attachment A: Asset list (WIN-10130, WIN-10054, WIN-10131, WIN-10011)
- Attachment B: OEM catalog Rev. Jan-2026
- Attachment C: Warranty terms (initial 5-year)
- Attachment D: VTM-SIEMENS-GAMESA-4-2 (reference)
- Attachment E: Addendum A (procedural conflict acknowledgment)

Cross-References:

- VTM-SIEMENS-GAMESA-4-2 (governing OEM manual)
- Standard MS-2025-03 (conflicting Operator standard — Addendum A)
- MOC-2025-022 (pending threshold-override request)
- VRA-2025-002 (Apex emergency subcontract pathway)
- OIR-2025-001, OIR-2025-003, OIR-2025-007 (asset-specific outage histories)

- RAM-2025-005, RAM-2025-007 (open replacement memos for WIN-10131, WIN-10011)
- RCA-2025-004 (root-cause analysis of repeat failures on WIN-10131)